

KAUSHIK PARVATHANENI

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Professional Summary

AI Application Engineer with nearly 3 years of experience building enterprise-grade AI systems using OpenAI, Azure OpenAI, Claude, and HuggingFace. Delivered end-to-end LLM-powered applications including RAG pipelines, multi-step reasoning agents, and prompt engineering frameworks integrated into enterprise insurance workflows on AWS and Azure. Proven ability to design backend AI services, automate ML pipelines, and connect LLM outputs to enterprise systems with direct domain experience in insurance, claims processing, and regulatory automation.

Technical Skills

AI Application Development: LangChain, LangGraph, OpenAI API, Azure OpenAI, Azure AI Services, Claude API, HuggingFace Transformers, Prompt Engineering, Multi-step Reasoning, Agentic Workflows

RAG & Retrieval: FAISS, RAG Pipeline Design, Chunking Strategies, Sentence Transformers, Embedding Models

Backend & API Services: FastAPI, Flask, REST APIs, Docker, CI/CD, AWS Lambda, AWS SageMaker

ML & Deep Learning: Scikit-learn, PyTorch, XGBoost, Fine-Tuning, ONNX, TensorRT, NLP, Feature Engineering

Programming: Python (Pandas, NumPy), SQL (CTEs, Window Functions), R, PySpark

Cloud & Data: AWS (S3, Lambda, Glue, Redshift, CloudWatch), Azure (OpenAI, AI Services), Snowflake, PostgreSQL, MongoDB, MLflow, Git

Work Experience

AI Application Engineer

August 2024 – November 2025

CMR Solutions LLC (Client: Blue Cross Blue Shield Association)

Richmond, USA

- Contributed to building a RAG pipeline using LangChain and FAISS over 40,000+ clinical policy documents, enabling natural-language querying that replaced a 2-hour manual search process and served 200+ daily internal users.
- Supported development of multi-step reasoning agents using LangGraph integrating OpenAI GPT-4 and Azure OpenAI endpoints, automating structured data extraction from unstructured insurance claims notes with 91% field-level accuracy.
- Worked on integrating AI capabilities into enterprise claims processing workflows by connecting LLM outputs to internal policy management systems via REST APIs, enabling automated decision support at scale.
- Assisted in implementing FastAPI backend services on AWS SageMaker to expose AI model endpoints; supported CloudWatch monitoring setup and contributed to rollback procedures during model updates.
- Participated in fine-tuning a HuggingFace classification model on labeled claims records to predict denial likelihood, contributing to a workflow that reduced the denial rate by 14%.
- Contributed to prompt engineering across enterprise use cases, evaluating prompt templates against held-out datasets to ensure consistent outputs across varied input formats.
- Built Python preprocessing pipelines to clean and chunk source documents before FAISS indexing, helping reduce RAG retrieval latency from 4.2 seconds to 1.8 seconds.

Data Scientist

May 2023 – April 2024

Kelly School of Business (FADS) – Indiana University

Bloomington, USA

- Built ML pipelines in Python using Apriori-based Market Basket Analysis and customer segmentation models to identify purchase patterns; outputs were integrated into marketing campaign workflows to drive targeted cross-sell strategies.
- Developed statistical pricing models using regression analysis and hypothesis testing in R; implemented a data validation framework that improved dataset accuracy by 20% and supported faculty-led strategic research.
- Automated data extraction and transformation pipelines from AWS S3 and Redshift using Python and SQL, feeding cleaned datasets into ML model workflows and reducing manual data preparation effort for downstream analysis.

Data Scientist

November 2021 – March 2022

Plotmydata

Hyderabad, India

- Trained and deployed classification and regression models using Scikit-learn on eCommerce datasets achieving 95%+ accuracy; packaged models as Flask REST APIs integrated into enterprise client systems.
- Performed feature engineering and data preprocessing pipelines in Python to clean, transform, and prepare structured datasets for model training, improving model input quality and reducing training time.
- Built Plotly-Dash dashboards displaying model predictions with confidence scores, enabling clients to review outputs and route low-confidence records for manual inspection within existing enterprise workflows.

Projects

Sentiment Analysis & Text Summarization App | Python, Streamlit, NLP, TextBlob, Gensim, Sumy, Pandas

- Built and deployed an end-to-end NLP web application for real-time and batch sentiment analysis of movie reviews, product reviews, and tweets; implemented text preprocessing, polarity and subjectivity scoring, and extractive text summarization using Gensim and Sumy.
- Deployed on Streamlit Cloud with a clean production-ready UI; enabled users to customize analysis parameters and download results — demonstrating end-to-end AI application development from NLP pipeline to live deployment.

AppFusion – App Recommendation System | Python, Streamlit, Pandas, SQL, SQLite, Plotly

- Built a personalized app recommendation system over 1.2 million Apple App Store records using data-driven filtering on attributes including genre, content rating, and user preferences; designed and queried a structured SQLite database to serve recommendations at scale.
- Deployed an interactive Streamlit dashboard with Plotly visualizations enabling users to specify preferences in detail and receive tailored app suggestions — demonstrating AI-powered personalization integrated with a backend data layer serving real user queries.

Education

Master of Science, Data Science	August 2022 – May 2024
Indiana University	Bloomington, USA
Bachelor of Technology, Computer Science	June 2018 – May 2022
GITAM University	Hyderabad, India

Certifications

AWS Certified Data Engineer – Associate	[Certification Link]
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